BRIJESH NANDA BRUNDA YOGANANDA

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PROFESSIONAL SUMMARY

Full Stack Developer and AI Engineer with nearly 4 years of industry experience engineering scalable microservices, now leading software and ML/AI engineering for a 15-person NSF AI Institute (ICICLE) research group at The Ohio State University. Builds production ML platforms spanning foundation-model-based auto-labeling, no-code workflow orchestration, and self-managed AWS HPC infrastructure, bridging research and production-grade software using Java, Python, and AWS.

EDUCATION

The Ohio State University, Columbus, OH

Master of Science in Computer Science (GPA: 3.94) | May 2026

Focus: Machine Learning, Computer Vision, MLOps, Advanced Algorithms, Data Mining and Visualization

The National Institute of Engineering, Mysuru, India

Bachelor of Engineering in Mechanical Engineering (GPA: 9.52) | Aug. 2016 – June 2020

Activity: Suspension Design Head, SAE BAJA India (built all-terrain vehicles)

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript, SQL, C++
- **AI/ML:** PyTorch, OpenCV, YOLO, HuggingFace, SAM, DINOv3, BioCLIP, OWLv2, CNNs, Transformers
- **Backend:** Spring Boot, Spring Security, Spring Data JPA, Maven, JUnit, Mockito
- **Cloud:** AWS (Lambda, EC2, ECS, S3, SQS, SNS, Kinesis Data Streams, IAM)
- **Frontend:** Angular, React, HTML5, CSS3
- **Databases:** MongoDB, MySQL, PostgreSQL

- **Tools:** Docker, Kubernetes, Jenkins, Git, Terraform, Linux, Kafka

EXPERIENCE

Associate Applications Developer (ICICLE) | The Ohio State University, Columbus, OH | June 2026 – Present

- Serve as the primary point of contact for software and ML/AI engineering within the ICICLE research group (8 undergraduates, 3 master's students, 4 PhD students), delivering tools and services to researchers across multiple domains and end users.
- Lead **Smart Labeler**, an interactive ground-truth generation platform for zero- and few-shot object detection, semantic segmentation, and video segmentation/tracking, combining SAM3 and OWLv2 for proposal generation with DINOv3 and BioCLIP for feature extraction in a human-in-the-loop triage workflow; reached ~80% accuracy after five feedback batches on animal ecology datasets, with few-shot batch jobs submitted to Slurm; extended to digital agriculture and weld-fault segmentation/tracking.
- Architect the **Workflow Orchestrator**, a no-code, open-source platform spanning the full ML/AI lifecycle (data collection, pre-processing, training, inference, harmonization, visualization, post-processing) with swappable, community-contributable steps featuring input/output ports, configurable triggers, and data validators; workflows bundle and run on laptops, base stations, or HPC clusters, with long-running steps deployed via Slurm-scheduled VM instances across multiple clusters using Apptainer and TAPIS integrations, including San Diego Supercomputer Center and OSC's Pitzer, Cardinal, and Ascend systems under user allocation IDs.
  - Built **Image Lab**, a no-code, OpenCV-based interactive step for image pre-processing, verification, and region-of-interest enhancement with batch job submission across directories.
  - Delivered an **AgriTech precision zone-spraying pipeline** on the Orchestrator: automated drone image upload triggers weed detection, generating a geopackage of spray/no-spray zones and a flight plan exportable to .waypoints, .rds (RD Pilot), and .plan (PX4) formats; built a companion Linux ground-control app for defining farm boundaries and obstacles and auto-generating flight plans from user parameters.
- Designed and built a custom **AWS ParallelCluster** HPC environment for the lab from scratch (head node, compute node groups with A10, L40S, and A100 GPU and CPU partitions), managing service quota increases, scoped queues and user groups, TAPIS

integration for external job submission, and attached Amazon EFS for private per-user storage; hold admin access and handle ongoing debugging and maintenance.
- Lead maintenance of the lab's websites, delegating minor updates to undergraduate and master's student contributors.

Graduate Research Assistant (ICICLE) | The Ohio State University, Columbus, OH | June 2024 – June 2026

- Architected an end-to-end ML pipeline that lets researchers run object detection without coding, pairing a drag-and-drop interface with SAM3 for one-click and text-prompt segmentation-assisted annotation.
- Implemented few-shot and zero-shot learning with OWLv2 and DINOv3, enabling rapid annotation of new object classes in animal ecology and digital agriculture with minimal training data.
- Built threshold optimization tools that visualize and tune parameters such as patch size against IoU score and similarity.

Software Development Engineer | Avis Budget Group, Bengaluru, India | July 2023 – July 2024

- Designed RESTful microservices processing 10,000+ daily transactions across fleet management, insurance, and payments, creating a unified rental operations ecosystem.
- Developed a data pipeline to manage toll transactions and streamline communication between fleet vehicles and toll systems.

Software Engineering Specialist | Dassault Systèmes, Bengaluru, India | Oct. 2020 – June 2023

- Engineered an Enterprise Message Bus on AWS (Kinesis, SQS, DynamoDB) processing 1M+ messages daily across 50+ microservices.
- Developed a customizable, web-based no-code workflow framework for designing backend systems, cutting the development lifecycle by 60%.
- Built a HIPAA-compliant Donor Management System handling 10,000+ records, digitizing donor onboarding, sample tracking, inventory control, and automated payments.

PUBLICATIONS

- **Resolve, Clear, or Defer: Training-Free Active Prototype Triage for Ecological Detection Pipelines.** WACV 2027, Applications Track. Accepted.
- **The Workflow Is the Curriculum: Provenance-Aware Learning Modules from AI/HPC Artifacts.** EduHiPC 2026, held in conjunction with the IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC 2026). Accepted.